

FINAL PROJECT DELIVERABLES — ACADÉMIE LIBRE (ACADEMIE_LIBRE_2025 LMS)

Completed: November 2025

Project Type: Full-Stack Custom Learning Management System (LMS)

Status: 100% Functional • Secure • Responsive • Production-Ready

1. Sprint 3 — Final Common Deliverables

By the end of Sprint 3, we transformed our early prototype into a complete, polished, and stable **Learning Management System**, fully customized for *candidats libres* preparing for national exams in Niger.

During this final sprint, we:

- Fully connected all frontend and backend pages
- Finalized the database with clean, normalized relationships
- Implemented secure login, registration, and session logic
- Integrated assignment submission, grading, quiz-taking, and module unlocking
- Polished the UI for both students and admins
- Ensured the entire system runs without errors or broken links
- Cleaned all folder paths, including routing issues
- Ensured the platform works smoothly on all screen sizes

This sprint marked the moment when the platform became **fully functional and demo ready**.

2. Final Architecture & Platform Structure (November 2025)

Frontend Architecture (PHP + HTML + CSS + Minimal JS)

The front end was designed to be simple, lightweight, and responsive. We used:

- **Pure PHP** for rendering dynamic content
- **HTML/CSS** for structure and design
- **Minimal JavaScript** for interactivity and module unlocking
- A clean, professional interface with:
 - Blue theme for students
 - Purple/dark theme for admins

Student-Facing Pages

- `dashboard.php` — student homepage
- `course.php` — list of available courses
- `module-view.php` — lectures + assignments + quizzes
- `resources.php` — course materials (PDF, links, videos)
- `assignments.php` — submissions and feedback
- `profile.php` — editable profile with photo upload
- `take-quiz.php` — quiz interface

Admin-Facing Pages

- `admin/dashboard.php` — admin overview + stats
- `admin/add_courses.php` — create/edit courses
- `admin/add_modules.php` — create/edit modules
- `admin/add_lecture.php` — create/edit lecture
- `admin/add_resources.php` — upload PDFs, links, books
- `admin/view_assignments.php` — grade student work
- `admin/post_assignments.php` — post student work
- `admin/add_quizzes.php` — manage quizzes and questions

Backend Architecture (PHP + MySQL)

We used a clean and consistent structure that resembles MVC:

- **`config/db.php`** — central PDO connection
- **`includes/functions.php`** — role checks, utilities
- **`upload-assignment.php`** — handles student submissions
- **`mark-complete.php`** — updates student module progress
- **`logout.php`** — clears session and redirects

Security Features:

- Prepared statements (PDO)
- Enforced session validation
- `is_admin()` and `is_logged_in()` role guards

- File upload filtering for safety

Database Architecture

The database was fully normalized and structured to support course delivery, progress tracking, assessments, and resource management.

Main Tables

users

courses

modules

lectures

user_progress

module_assignments

assignments (student submissions)

course_resources

quizzes

questions

quiz_attempts

Each table is linked through proper foreign keys to support:

- sequential module unlocking
- assignment submission and grading
- resource linking to courses
- user progress tracking
- quiz attempts and scoring

Folder Structure (Descriptive Overview)

C:\xampp\htdocs\Academie_Libre_2025\

| index.php

| login.php

- | register.php
- | logout.php
- | dashboard.php
- | course.php
- | module-view.php
- | resources.php
- | assignments.php
- | profile.php
- | take-quiz.php
- | upload-assignment.php
- | mark-complete.php
- |
- |— admin/
- | dashboard.php
- | courses.php
- | modules.php
- | lectures.php
- | quizzes.php
- | add_resources.php
- | view_assignments.php
- |
- |— assets/
- | css/style.css
- | uploads/profile/
- | uploads/assignments/
- | uploads/corrected/
- | uploads/resources/

```
|
|
|— includes/
|   header.php
|   footer.php
|   admin-header.php
|   functions.php
|
|— config/
|   db.php
```

3. User Flow

Student Journey

1. Login using credentials
2. Enter dashboard
3. Select course
4. Open module
5. Watch lectures (each one onlyb unlocks when the previous is completed)
6. Mark lecture as complete
7. Submit assignment (PDF/DOCX/ZIP)
8. Take quiz
9. View resources
10. Receive grades and feedback
11. Complete all modules → certificate (future feature)

Admin Journey

1. Login via admin portal
2. Access dashboard for stats and quick links
3. Create or edit courses

4. Add modules and lectures
5. Upload resources
6. Post assignments
7. Grade submissions and return corrected files
8. Track student progress
9. Manage quizzes

4. Core Features

The platform includes a rich set of core features that were fully developed and tested during this sprint.

Secure User Authentication

We implemented a complete login and registration system with fully protected sessions. The platform automatically detects whether the user is a student or an admin and redirects them accordingly. All session data is sanitized and validated to prevent misuse.

Dynamic Student and Admin Dashboards

Students see their enrolled courses, modules, and progress, while admins access tools such as course creation, resource management, and assignment review. Each dashboard is visually distinct and themed for clarity.

Complete Course–Module–Lecture System

Courses contain modules, and each module contains lectures, assignments, and quizzes. Lectures are displayed using YouTube embeds pulled dynamically from the database.

Modules unlock sequentially, meaning students can only access the next lecture or module after marking the current one as complete.

Smart Progress Tracking

Every time a student finishes a lecture or module, the progress table updates automatically. This ensures a structured learning path and encourages students to move step-by-step through the course.

Assignment Management System

Admins can post assignments, including instructions, reference files, and deadlines. Students can upload PDF, DOCX, or ZIP files, which are stored in dedicated folders. The system ensures proper file handling and displays uploaded submissions instantly.

Grading and Feedback Workflow

Admins can download assignments, upload corrected files, add textual feedback, and record grades. Students instantly see:

- their submitted file
- the grade assigned
- admin notes or comments
- the corrected version of their submission

Resource Management

The admin can upload different types of resources such as:

- PDF notes
- external links
- textbooks
- YouTube videos

Students see resources filtered by course and can preview files directly in the browser.

Interactive Quiz System

The quiz interface allows students to answer multiple-choice questions. Results are saved to the database, and admins can view attempts through the quiz management panel.

Profile Management with Photo Upload

Students can update their personal information and upload a profile picture. The system stores images in structured folders and immediately updates the UI.

Professional, Responsive UI

The entire platform is optimized for smooth navigation, clear visuals, and ease of use. The themes (blue for students, purple for admins) create a distinct identity and improve the user experience. Every screen scales well on both mobile and desktop.

5. Remaining / Future Features

The following improvements are planned for the next version:

- AI-powered evaluation for assignments
- Real-time chat system
- Automated email notifications
- Mobile app version (Flutter)

- Auto-generated completion certificates

FINAL PROJECT DELIVERABLES — INDIVIDUAL REPORTS

1. Foureiratou — Individual Contribution Report

A. Role

My primary responsibilities focused on **content development**, **resource curation**, and shaping the **educational flow** of the platform. I gathered learning materials, created full courses, designed quizzes and assignments, and uploaded all academic PDFs. I also worked on the **ER diagram**, participated in **database setup**, and contributed heavily to the **project documentation** and **problem definition**.

B. Features and Tasks Completed

- Gathered and organized all academic resources
- Created complete courses and structured weekly modules
- Designed quizzes and assignments for each course
- Uploaded all learning PDFs and external resources
- Developed the ER diagram used for database implementation
- Supported database setup, testing, and hosting
- Ensured the curriculum followed a logical sequence for *candidats libres*
- Assisted with UX planning to match course flow and user needs
- Contributed to documentation, research, and overall platform planning

C. Key Contribution

My largest contribution was designing the full learning structure, the weekly modules, course sequencing, quiz placement, and assignment timelines. This defined the academic workflow that guided all development decisions.

D. Challenges

- Structuring materials to match Niger's free-candidate system
- Converting offline curriculum into digital weekly lessons
- Managing a large volume of PDFs and resources
- Balancing content creation with technical tasks
- Learning hosting and database migration processes

E. Lessons Learned

- Strong content architecture makes technical implementation easier
- Accurate user understanding improves platform relevance
- Documentation and planning are essential for team coordination
- Database setup requires consistent naming and structure
- Hosting and deployment demand patience and repeat testing

2. Chidima — Individual Contribution Report

A. Role & Main Contribution

Chidima served as the **main frontend and backend programmer** of the team. She built most of the PHP pages, implemented the UI design, handled styling, and translated platform ideas into functional, interactive pages.

B. Features Completed

- Built major platform pages: Dashboard, Progress, Admin, and parts of Resources
- Created responsive, consistent styling using CSS
- Added progress bars and dynamic UI indicators
- Implemented login, sessions, and basic backend logic
- Added weekly **locking/unlocking** progression
- Developed smooth page navigation and routing
- Structured resource pages with videos, notes, links, and PDFs
- Ensured UI consistency across all user pages

C. Key Technical Design

Her major contribution was creating the sequential learning progression logic system that unlocks Week 2 only after Week 1 is completed. This supports structured and disciplined learning.

D. Challenges

- Maintaining consistent UI across multiple pages
- Ensuring responsiveness for various devices
- Testing weekly progression logic

- Handling large, multi-page CSS structures

E. Lessons Learned

- Modular CSS makes large projects easier to manage
- User experience depends on clear navigation and flow
- Pre-planning UI layouts shortens development time
- Early testing prevents routing and linkage errors

3. Alan — Individual Contribution Report

A. Role & Main Contribution

Alan handled **the MySQL database development** and the design of the **typical user flow**. He was not involved in front-end or back-end programming aside from database structure support.

B. Features Completed

- Developed all database tables using the ER Diagram
- Designed relationships for students, subjects, quizzes, and assignments
- Implemented login and registration database logic
- Created test_connection.php for backend testing
- Supported backend needs with correctly structured tables

C. Key Technical Design

Alan designed the critical database relationships:

- user_subjects linking students to their chosen subjects
- quizzes → questions → quiz_results
- homework → homework_submissions
- resources connected to subjects and weekly modules

These structures support progression, grading, and resource delivery.

D. Challenges

- Designing correct foreign-key relationships
- Ensuring compatibility with frontend data requirements
- Synchronizing database setup with team expectations

E. Lessons Learned

- Good database planning removes future system errors
- Testing backend connections early prevents deployment issues
- Documentation improves collaboration on backend logic
- Consistent naming conventions are critical for stability

5. Important Links

GitHub Repository

<https://github.com/Furairah3/Webtech-Academie-Libre-Projects.git>

Live Server (if available)

http://169.239.251.102:341/~foureiratou.idi/Academie_Libre/Webtech

CMS / WordPress

<https://fzakariyaouidi.wixsite.com/academie-libre>

Local Development Links

- Student Portal: http://localhost/Academie_Libre_2025/
- Admin Portal: http://localhost/Academie_Libre_2025/admin/
- Project Directory: C:\xampp\htdocs\Academie_Libre_2025\

6. Conclusion

Académie Libre is a complete, stable, and fully functional LMS designed specifically for the realities of *candidats libres* in Niger.

The system supports structured learning, sequential progression, resource management, assignment workflows, and administrative control.

This project demonstrates:

- strong teamwork
- deep full-stack development skill
- thoughtful educational design
- real-world problem solving
- and a meaningful contribution to accessible education technology.

